

Mechanized Excavation Techniques

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The Excavation Process

It is instructive to consider the fundamental objective of the excavation process - which is to remove rock material (either to create an opening or to obtain material for its inherent value). In order to remove part of a rock mass, it is necessary to induce additional fracturing and fragmentation of the rock.

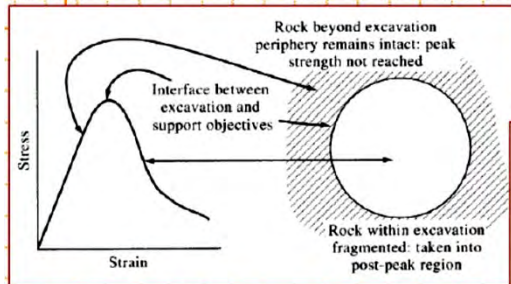
This introduces three critical aspects of excavation:

- ➡ The peak strength of the rock must be exceeded.
- ➡ The *in situ* block size distribution must be changed to the required fragment size distribution.
- ➡ By what means should the required energy be introduced into the rock?



Energy and Excavation Process

Although the objective during excavation is **large-scale fragmentation**, at the same time we wish to **minimize any damage** to the wall rock as this would work towards weakening the rock mass which may result in ground control problems. Fragmentation and rock mass damage are both related to the amount of energy used and whether its applied instantaneously or continuously.



Hudson & Harrison (1997)

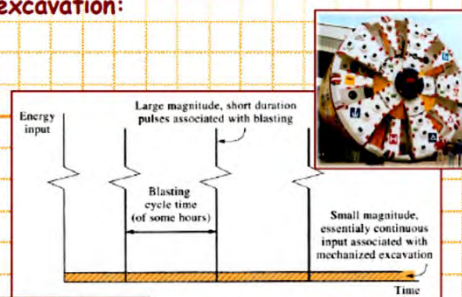
The tensile strength of rock is about 1/10th the compressive strength and the energy beneath the stress-strain curve is roughly its square. Therefore, breaking the rock in tension requires only 1/100th of the energy as that in compression.

Energy and Excavation Process

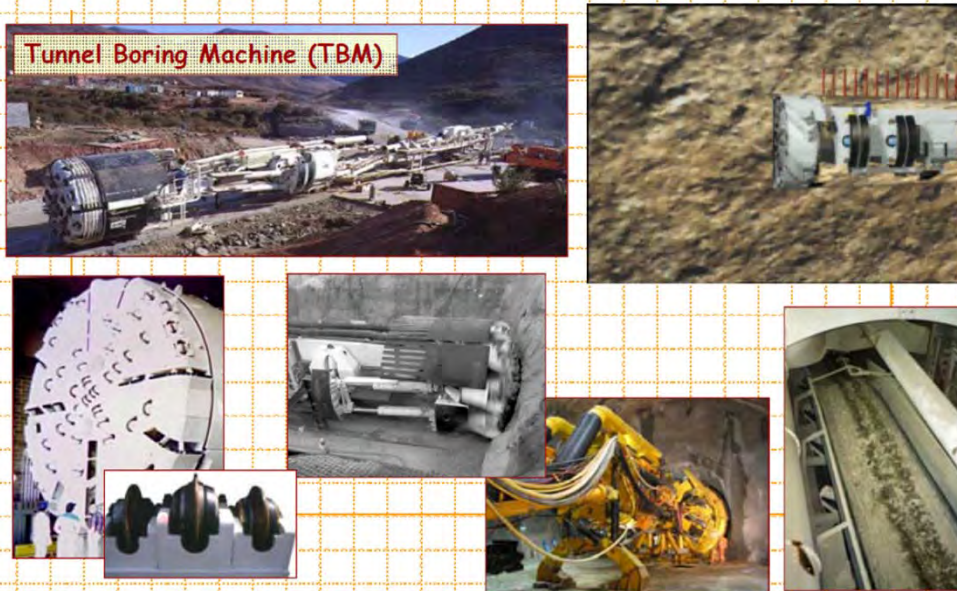
One objective in the excavation process may be to optimize the use of energy, i.e. the amount of energy required to remove a unit volume of rock (**specific energy** = J/m^3). There are two fundamental ways of inputting energy into the rock for excavation:

➡ **Blasting:** Energy is input in large quantities over very short durations (cyclical - drill then blast, drill then blast, etc.).

➡ **Machine Excavation:** Energy is input in smaller quantities continuously.



Machine Excavation in Rock



Machine Excavation in Rock

There are two basic types of machine for underground rock excavation:



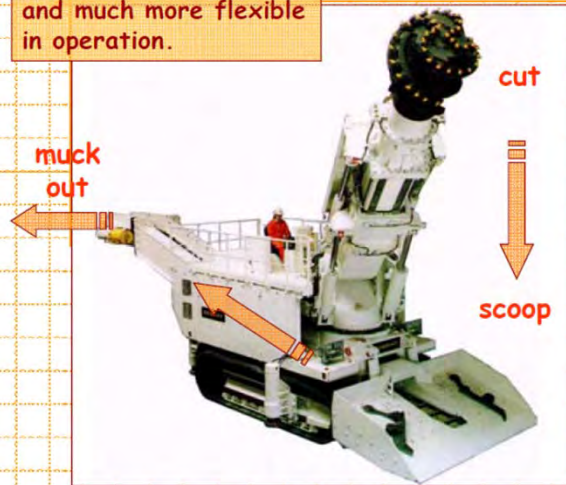
Partial-face machines: use a cutting head on the end of a movable boom (that itself may be track mounted).



Full-face machines: use a rotating head armed with cutters, which fills the tunnel cross-section completely, and thus almost always excavates circular tunnels.

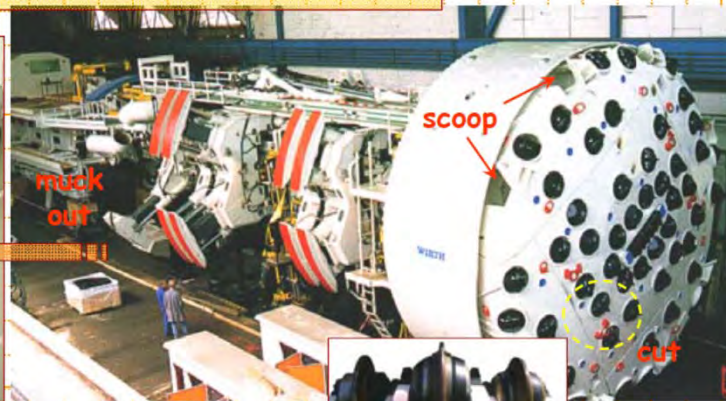
Machine Excavation in Rock

Partial-face machines are cheaper, smaller and much more flexible in operation.



Machine Excavation in Rock

Full-face machines - when used for relatively straight and long tunnels (>2 km) - permit high rates of advance in a smooth, automated construction operation.



Machine Excavation

The advance rate at which the excavation proceeds is a function of the **cutting rate and utilization factor** (which is the amount of time that the machine is cutting rock). Factors contributing to low utilization rates are difficulties with ground support and steering, the need to frequently replace cutters, blocked scoops, broken conveyors, etc.



The cutters may damage if the TBM is pushed forwards with too much force, or large blocks fall and strike them.



Broken conveyor

Classification of Underground mechanical excavator by rock type

Hard rock	Soft rock
1.Roadheader	1.Roadheader
2.TBM	2.Continuous Miner
3.Mobile Miner	3.Shearer
4.Raise Borer	

Comparison of mechanical excavation with drill and blast systems

Advantages	Disadvantages
• Generally gives more rapid advance than blasting in soft to medium-hard rock. • Fewer unit operations • Smoother excavation profiles • No blasting damage to surrounding rock • Inherently safer	• Cannot excavate very hard rock • High capital cost • Require specialist maintenance • Not as flexible

Parameters influencing applicability of mechanical miners

- The decision to introduce a mechanical miner is generally based on the predicted performance of the selected machine in the given geological and operational conditions.
- Since it is usually not possible to change the geological conditions, a thorough study of the geological and geotechnical parameters (rock mass and intact rock properties) at the feasibility and planning stages should be carried out to match the machine and operational parameters to the geological conditions.

Machine considerations

- The general technical requirements of excavation machines, in addition to safety and economy, are selective mining ability, flexibility, mobility and hard and abrasive rock-cutting ability.
- Selective mining ability is the ability to cut selectively in mixed face conditions so that the mineral can be excavated separately, reducing dilution.
- Mobility means easy relocation of machines from one face to another, when necessary.
- Flexibility means easy adaptability to changing operational conditions, such as face cross-section shape (horseshoe, rectangular, circular, etc.), gradient, turning radius and unevenness of the floor.

Machine considerations

- Hard and abrasive rock-cutting ability is the most important limiting factor on the performance of mechanical miners; applications show that strength, texture, etc., of the geological environment are the basic input parameters for the selection of mechanical miners and performance prediction. the most efficient type of tool in hard and abrasive rock.

Geological and geotechnical considerations

- Rock mass features (such as joint sets, bedding planes, foliation, hydro geological conditions, deposit geometry, etc.) and intact rock properties (such as cuttability, abrasiveness etc).

Operational considerations

- Mine layouts and development drivages are planned in accordance with the mining method, which is selected on the basis of orebody shape and dimensions and other deposit characteristics.
- For reliable, economic operation the characteristics and layout features of a mechanical miner should reflect whether it is to be used for the excavation of drifts or for ore/mineral. The layout might be modified for a particular mechanical miner if necessary.
- The back-up equipment (such as muck transportation, support, etc.) selected for a new mine must be chosen to be compatible with the mechanical miner. In an existing mining operation, however, the machine might be selected on the basis of the equipment already in use.

Main influencing parameters for excavation

Rock

- ☐ Rock composition
- ☐ Abrasivity (equivalent quartz content)
- ☐ Matrix
- ☐ Density (porosity/dry gross density)
- ☐ Strength: compressive-tensile-shear
- ☐ Destruction energy
- ☐ Post-failure behaviour

Main influencing parameters for excavation

Rock mass

- ☐ Joint structure: spacing, roughness, orientation
- ☐ Anisotropy: orientation, formation
- ☐ Primary stress conditions
- ☐ Weathering: type and degree
- ☐ Hydrothermal decomposition

Compressive strength

A stronger rock is more difficult to cut. Unconfined (also called *uniaxial*) compressive strength (UCS) is the most widely accepted parameter to measure strength of a rock sample, and is usually considered to be the most useful guide to the performance of mechanical excavation (Brown & Phillips, 1977).

Density

A rock with higher specific gravity will obviously need higher specific cutting energy (SE). The cutting performance of a machine in terms of bank volume cut per unit time in a coal seam with specific gravity of around 1.5 will definitely be higher than that of the same machine in other rocks with comparable UCS but specific gravity of about 2.5.

Abrasivity

More abrasive a rock, more it causes wear and tear on cutting tools of the machine negatively affecting its performance. Abrasiveness of a rock is expressed in terms different indices, like

Cerchar Index, Schimazek-F Index etc. that are determined by testing intact rock samples in the laboratory following standard procedures.

Discontinuities

Presence of joints and other structural features like bedding planes, cleats and slips etc. in high frequency, and their length and degree of openness assist the cutting process, especially when they are favorably oriented with respect to the direction of cutting.

Moisture content

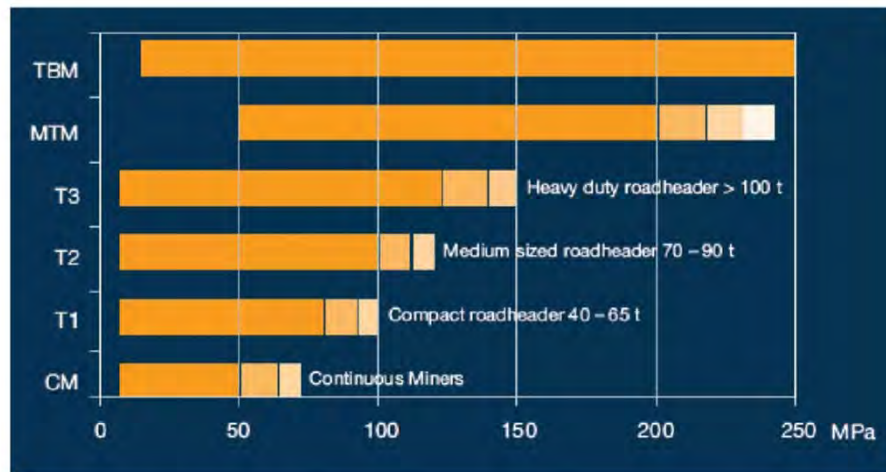
Mechanical strength of rock commonly reduces with an increase in moisture content. In a study it was observed that both cutting and normal forces decreased with moisture content by up to 40 and 49% respectively, SE reduced by 38%, and impact wear of the cutting tool dropped by 80% when cutting a saturated sandstone sample with 68% reduction in UCS compared to the dry sample (Mammen *et al*, 2009).

In situ stress condition

Absence of discontinuities and high *in situ* stress might critically increase the cutting resistance and may make cutting difficult. Presence of major geological features, like faults, shears, folds, igneous intrusions etc. may locally affect the *in situ* stress condition at any site.

Other Material properties

Assessing the cuttability of a rock formation with respect to a mechanical excavation system is an imprecise area of endeavor (Roxborough, 1987). Nonetheless, several researchers have developed certain empirical approaches to estimate cutting performance of machines for different rocks on the basis of a number of properties of rocks. Other than the parameters already mentioned, they include modulus of elasticity, point load index, Schmidt hammer rebound number, fracture energy, toughness index, Brazilian tensile strength, brittleness, sonic velocity, stickiness of material, specific cutting energy, various indices for joint characteristics, rock quality designation (RQD) etc. of rocks.



CM- Continuous Miner, MTM- Mobile Tunnel Miner, TBM- Tunnel Boring Machine

Road Header:

- Roadheaders are the most widely used underground partial-face excavation machines for low to medium strength rocks.
- They are used for both development and production in soft rock mining industry (i.e. main haulage drifts, roadways, cross-cuts, etc.) particularly in coal, sedimentary rocks, industrial minerals and evaporitic rocks.
- In civil construction, they find extensive use for excavation of tunnels (railway, roadway, sewer, diversion tunnels, etc.) in soft ground conditions, as well as for enlargement and rehabilitation of various underground structures.
- Their ability to excavate almost any profile opening also makes them very attractive to those mining and civil construction projects where various opening sizes and profiles need to be constructed.
- **"Roadheading"** machines with a cutter head designed to excavate stone rather than coal instead of drilling and blasting.
- This gives a more continuous process and should give a good drift profile.
- The use is limited by the grades involved (maximum 1 in 4 (14°)) and the hardness of the material to be cut (maximum 60-150 MPa UCS).

- In addition to their high mobility and versatility, roadheaders are generally low capital cost systems compared to the most other mechanical excavators.
- Because of higher cutting power density due to a small cutting drum, they offer the capability to excavate rocks harder and more abrasive than their counterparts, such as the continuous miners and borers.
- Roadheaders were first developed for mechanical excavation of coal in the early 50s.
- The major improvements achieved in the last 50 years consist of steadily increased machine weight, size and cutterhead power, improved design of boom, muck pick up and loading system, more efficient cutterhead design, metallurgical developments in cutting bits, advances in hydraulic and electrical systems and more widespread use of automation and remote control features.
- All these have led to drastic enhancements in machine cutting capabilities, system availability and the service life.

- Machine weights have reached from 17 tons up to 120 tons providing more stable and stiffer (less vibration, less maintenance) platforms from which higher thrust forces can be generated for attacking harder rock formations.
- The cutterhead power has increased significantly from 37 kW approaching to 500 kW to allow for higher torque capacities. Modern machines have the ability to cut cross-sections over 100 m² from a stationary point.

In **comparison to drill and blast methods**, the main **advantages** of roadheaders are:

- ❑ One machine is capable of cutting, loading and assisting in the erection of supports.
- ❑ Compared with conventional driving, a greater advance per manshift is obtained.
- ❑ Roof control is improved since the roof and sides of the roadway are not shattered by shotfiring.
- ❑ The machine is very stable, easily controlled and capable of cutting a large variety of roadway sections and sizes.
- ❑ It is safer since the men are near to the unsupported strata only during the erection of supports.

Mechanical Tunneling

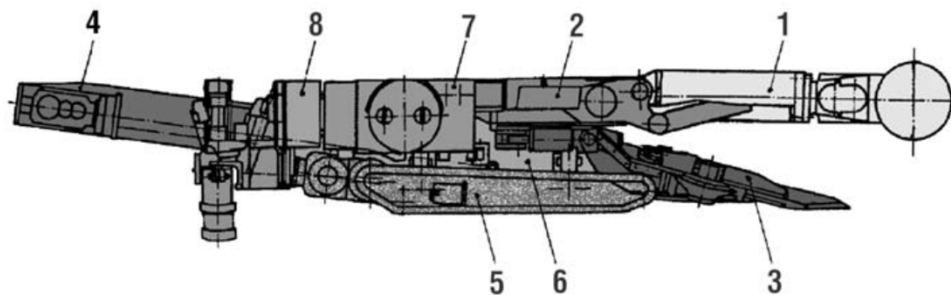
PART FACE

ROADHEADERS

Basic design and operating features

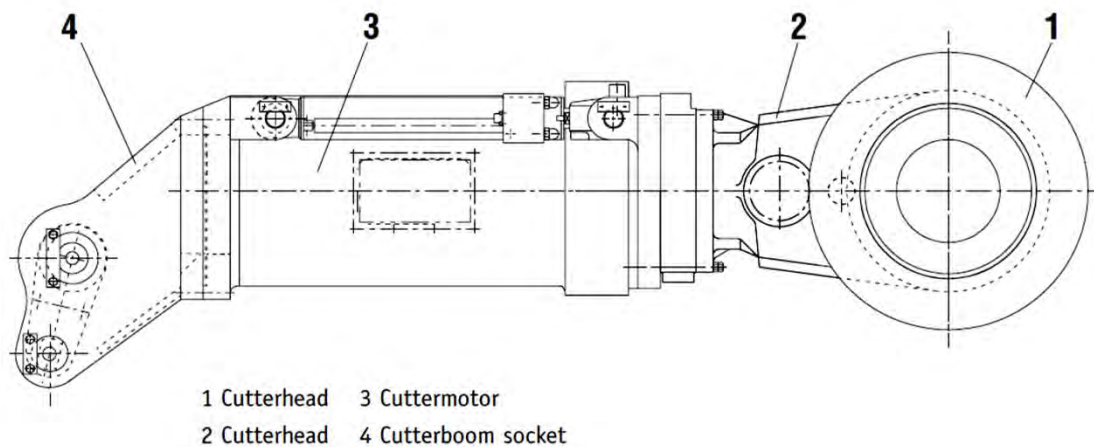
The standard roadheader features the following functions:

- Rock excavation (rock cutting)
- Gathering of excavated muck
- Muck transfer to secondary conveying equipment
- Machine transfer



- 1 Cutter Boom
- 2 Turret
- 3 Loading Assembly
- 4 Chain Conveyor
- 5 Track drive
- 6 Frame
- 7 Electric Equipment
- 8 Hydraulic Equipment

Roadheader main assembly groups



Cutter boom components

Basically a roadheader consists of a cutting unit, a gathering unit and a delivery unit. This equipment is very mobile and versatile compared with a full face TBM. It can cut a variety of cross sections; change diameters at will, change directions quickly, and move to and from a face under its own power. The machine usually incorporates gathering arms and a conveyor system to move the material cut from the face to a loading point at the rear of the machine. From this point, the muck may be handled by variety of methods available; and that includes shuttle trains, conveyors, or trucks.

Roadheader cutting boom is usually mounted on crawler track but increasingly the booms are being mounted on other machines such as: hydraulic breakers, trucks, traveling gentries and inside the shields.

Modern roadheaders are equipped with electronic/hydraulic-controlled systems linked to microprocessor-based guidance and profile control systems.

➤ Depending on cutting head, Roadheader are two types.

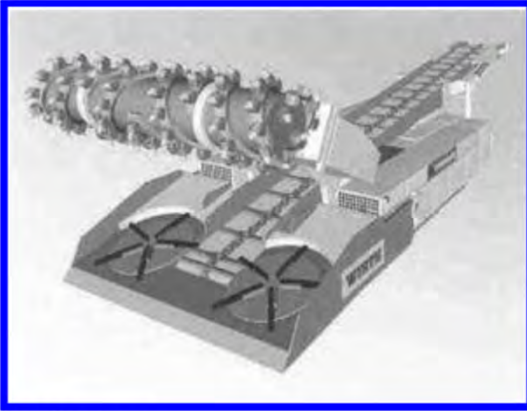
- Borer type-Transversal - rotates parallel to the cutter boom axis-DOSCO
- Ripper type-Longitudinal - rotates perpendicular to boom axis – AM-50

➤ It can work gradient flatter than 18 deg/ 1 in 3.

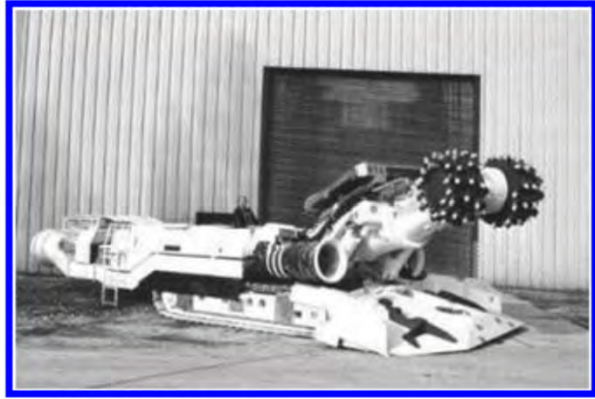
CLASSIFICATION BOOM MOUNTED ROADHEADERS

Based on the cutting principle employed, the roadheaders, or continuous miners can be classified as :

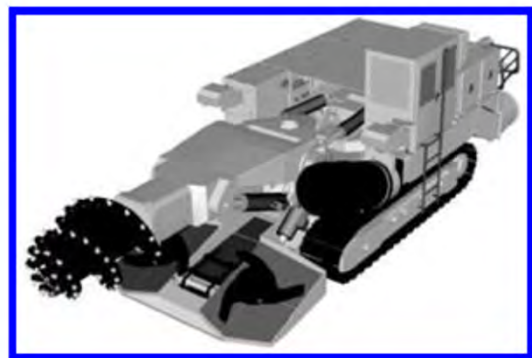
- Ripper or Transverse type: Bar and Disk
- Milling or longitudinal (Auger) type
- Borer type.



(a) Bar type



(b) Ripper type



(c) Milling type

Ripper (transverse) type Roadheaders – (cutter heads with rotation perpendicular to the boom axis):

In ripper type the full weight of the machine acts as the counter reaction for the cutter head. The rock is ripped off the face and thrown on the gathering head. Miner utilizes the weight of machine as the reaction force for cutting in a better way than the milling type of roadheaders, and no bracing jacks 'stells' are required. These machines have 20–35% lower weight (which means lower price) than boom type roadheaders of equivalent capacity.

Bar type Roadheaders:

Its cutting element consists of a ripper bar, or cutting head that tears the coal/rock from the face. The cut rock/coal is carried by the moving chains and discharged into an intermediate conveyor. The rock/coal that falls on the floor is picked by gathering arms and loaded on the conveyor. The cutting head consists of five to seven cutting chains with picks which runs in guides all around the ripper bar. The ripper bar is hinged at the rear end, which permits the front end to raise or lower. The chains are driven by electric motors. The whole unit i.e. the cutting chains, the motors and the conveyors can be telescoped forward to cut the rock/coal.

To give a cut, the machine is positioned in the center of the face with cutting head retracted. The machine's cutting head is swung to one side of the face and lowered down to the floor. With the chain in motion the ripper bar is hydraulically pushed forward in the rock /coal (usually 460 mm). After it has been fully advanced the ripper bar is gradually raised until it reaches up to the roof and this completes one strip. With a meter wide ripper bar, five such slices are required to complete a 5 m wide gallery (tunnel). After full face has been cut, the machine is moved forward.

Disc type Roadheaders:

In this type of ripper unit, there are two cutting heads, each consisting of two vertical discs laced with tipped bits. The cutting heads are carried in the front end of an extension boom. In this machine the cutting heads are rotated and at the same time a to-and-fro movement of about 25 times/minute is imparted to the gear case and the cutting heads. The resulting motion, therefore, of each pick is spiral which results in low consumption of power and less degradation of rock/coal. In one of the designs, for example, the cutting head cuts a kerf of 2.5 m wide and 0.6 m deep. The discs rotate at the speed of 75 r.p.m. The cutting head can be lowered or raised hydraulically. Apart from the cutting head, the machine has got an apron with gathering arms, a conveyor extending up to the rear of the machine and crawler tracks.

Using these heading machines it is possible to drive or construct large sized chambers or even the tunnels of large size can be driven in two lifts. The method is known as 'benching'. First upper bench is advanced in the upper half portion of the face and then the lower follows it.

Milling or Longitudinal (Augur) Roadheaders:

In milling type roadheaders a cylindrical or cone shaped cutter head rotates in line with the axis of the cutter boom. The cutting force is exerted mainly sideways, which prevents utilization of full weight as counter force. When cutting harder rock the machine is braced against the sidewalls with hydraulic jacks ('stelling'). This consumes time and the bracing jacks, who protrude side ways, make the machine inflexible in narrow headings. For wider and high tunnel faces, particularly in the hard rock, these types of headers are unsuitable because their bracing jacks (stells) cannot reach both sidewalls (ribs) and the roof to stabilize the roadheader.

These headers rip the rock from face and throw it side ways on the floor. These headers have small diameters than ripper heads and better suited for the selective mining of thin ore bands or lenses of high grade. At present the roadheaders with inter changeable cutter heads either ripper or milling (augur) are available.

Borer Type Roadheaders:

These types of roadheaders have been designed to cut the core of coal from the face by boring large diameter holes. A large amount of coal is obtained without being actually cut, therefore, the proportion of large sized coal is high. Numbers of designs of this type of headers are available with a variation in the design of cutting heads.

Coal from the face drops to the bottom and is pushed on a scraper chain conveyor by rotating arms, which are provided with ploughs. The scraper chain conveyor runs centrally through the machine and carries the coal to the rear end. The rear end can be swung on either side by 40° or can be raised or lowered hydraulically. The entire cutting unit is supported on hydraulic cylinders, which can tilt the cutting unit by 4 degree in the horizontal or vertical plane. This allows borer to negotiate the irregularities of the seam. This machine can drive the gallery of 1.8 to 2.3 m high and 3.2 to 3.8 m wide while advancing at the rate of 0.3 to 0.75 m/minute.

Classification of Boom Type road Header:

Generation:

Generation	Weight (Tonnes)	Compressive Strength of Rock to be cut (MPa)
1 st (1960)	9	40
2 nd (1970)	22 to 37	85
3 rd (1976)	45 to 70	100
4 th (2000)	120	100 to 160

Weight:

Tucker (1985) classified roadheaders according to weight as:

- Light Duty; weight up to 30 t, cutting capabilities up to 70 MPa
- Medium Duty; weight between 34 to 45 t, cutting capabilities up to 100 MPa
- Heavy Duty; weight over 45 t, cutting capabilities up to 150 MPa

Class	Weight
0	<20 t
I	20-30 t
II	30-50 t
III	50-75 t
IV	>75 t

Atlas Copco – Eickhoff established the following classification according to weight (Schneider, 1988);

Classification of Boom Type road Header:

Neil et. al (1994) refer road headers as small size up to 30 t, midsize between 30 to 70 t and large size between 70 to 120 t.

Cutting Action of heads:

- Milling type or Borer Type (axial head), and
- Ripping Type (Transverse head)



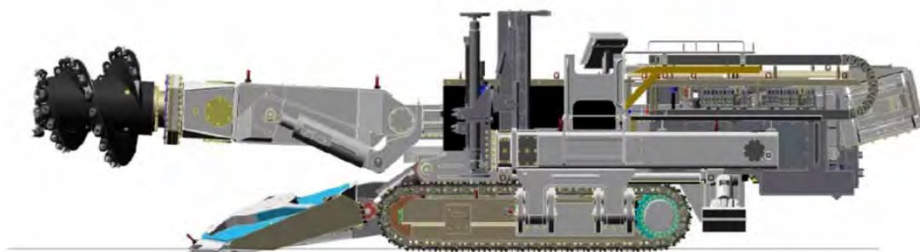
Fig: Cutting action of road headers
a. Ripping type, b. milling type



Borer type Roadheader



Borer type Roadheader



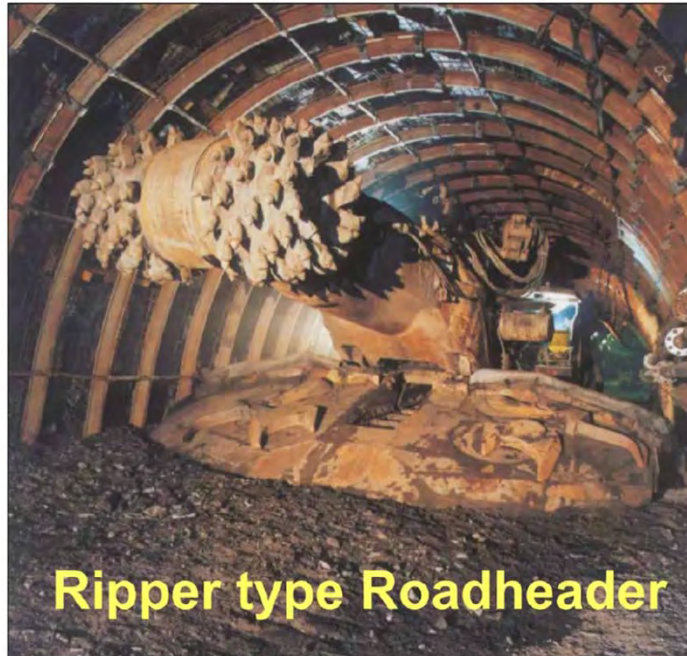
Borer type Roadheader (Side view)

Borer type Roadheader



Ripper type Roadheader



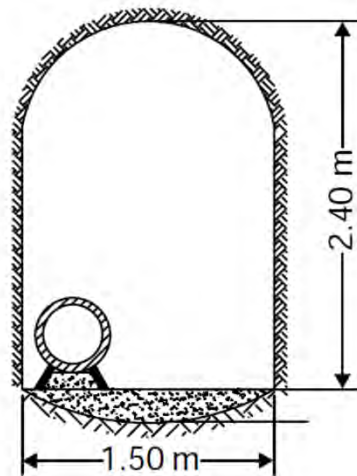


Ripper type Roadheader

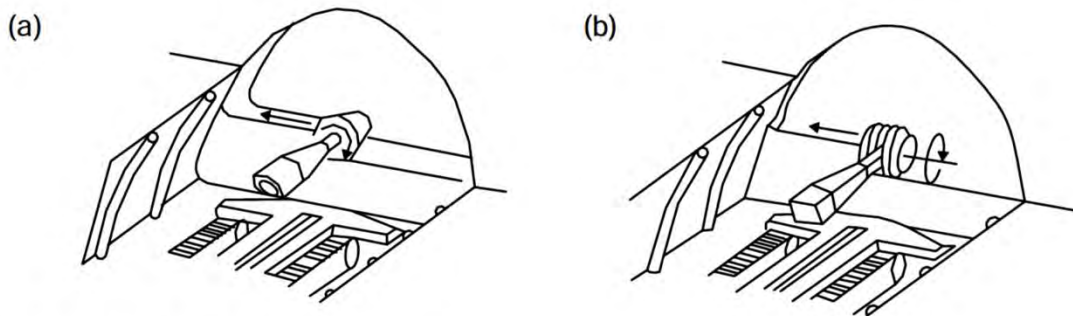
Cutting tools on Roadheader

- Radial picks are generally suitable for cutting soft to medium hard rocks and coal.
- Forward attack picks are also termed tangential picks, together with point attack picks due to the orientation of their tool axis. Such picks can also be used for cutting soft to medium hard rocks.
- Point attack picks, also known as pencil point tools, have been increasingly employed in medium to hard rock cutting and become an inevitable tool on medium and heavy duty roadheaders.



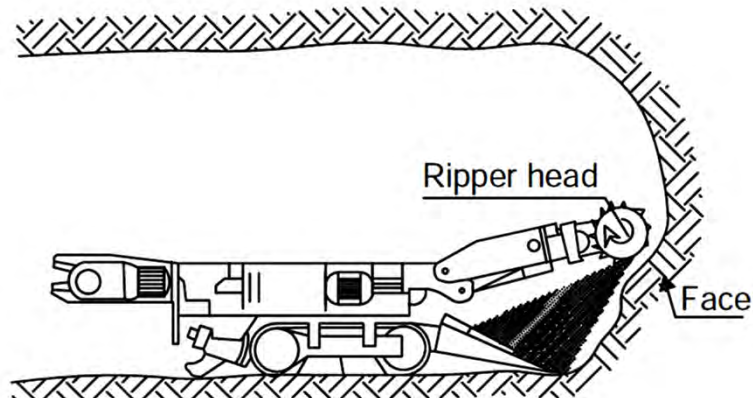


Opening's shape obtained by a partial face tunnel borer



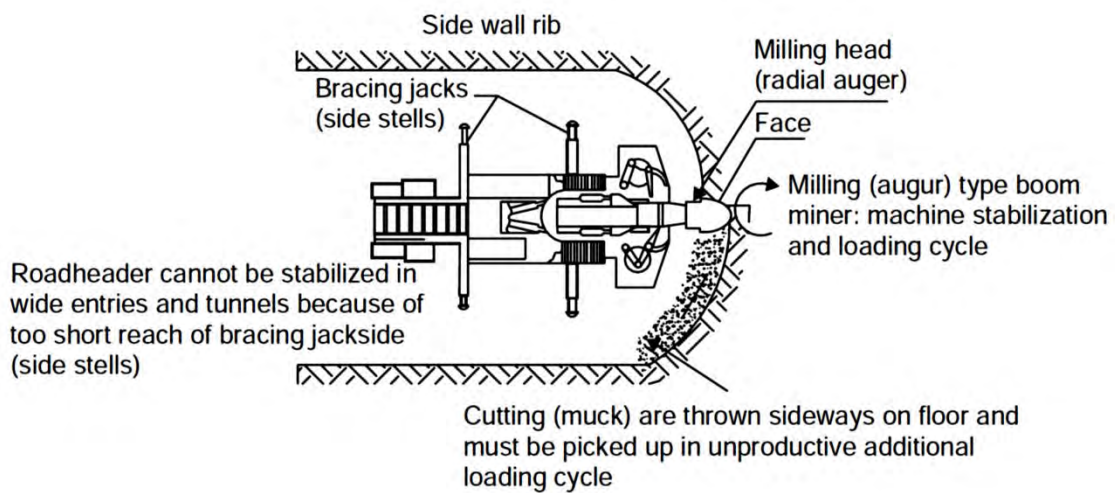
Comparison of two cutting systems:

- (a) Milling (auger) type cutter head can mine narrow bands and lenses selectively
- (b) Ripper type cutter head having 25–30% higher rate of production



Approx. 80% of cuttings are thrown to the gathering head

Ripper or transverse type roadheader



Milling/longitudinal roadheader

Table: Classification of roadheaders based on weight

RH – Roadheader.

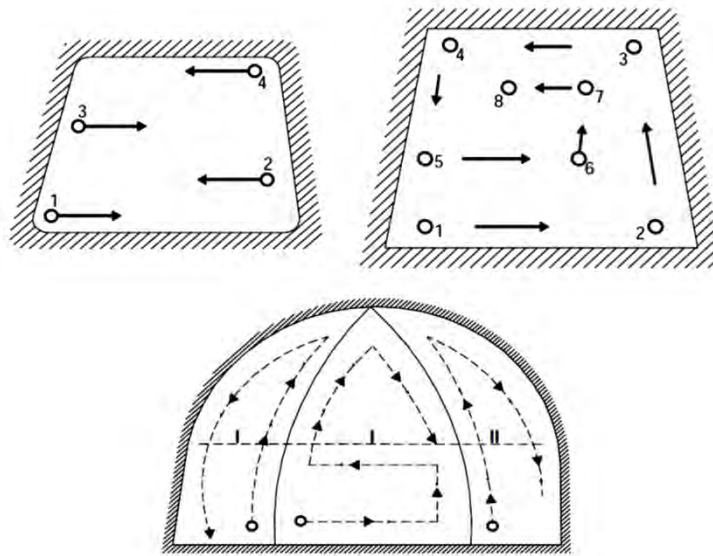
Roadheader	Weight range (t)	Cutter head power (kW)	RH with standard cutting range		RH with extended cutting range	
			Max. Section (m ²)	Max. UCS (MPa)	Max. Section (m ²)	Max. UCS (MPa)
Light	8–40	50–170	~25	60–80	~40	20–40
Medium	40–70	160–230	~30	80–100	~60	40–60
Heavy	70–110	250–300	~40	100–120	~70	50–70
Extra heavy	>100	350–400	~45	120–140	~80	80–110

ADVANTAGES OF ROADHEADERS

Versatility and Mobility: TBM generates circular openings whereas this unit can generate variety of sections. Face can be accessed, and by retraction a roadheader from the face; all required measures for rock protection (support) can be performed without significant shutdown; and this feature enable its use in the changing rock conditions. Large tunnels and openings can be subdivided and excavation activities can be carried simultaneously in these sections/divisions.

Low Investment: Compared to the TBM, for the similar size of cross-section; the investment costs for roadheaders amount to be approximately 0.15 (large sections) to 0.3 (small sections). Since they are available on rent basis, as such, they can be used even for small projects.

Quick and Easy Mobility: Comprehensive assembled equipment and chambers are not required. Delivery time is usually 3–6 months, and immediate after the delivery they can be brought into operation.



Working sequence of roadheaders. Left: Attacking from bottom to top.

Right: Attacking perimeter. Bottom: Attacking middle of face first then sides.

Procedure of driving by the heading machines:

The heading machines also operate in a cycle, whose duration is a function of length of unsupported roof, or the roof that can be allowed unsupported. It varies from face to face.

- Excavating the perimeter first and then the central part, if the strata are friable.
- Ripping bottom first then the upper part to take advantage of undercutting. It gives better advance/output rates.
- If different bands or layers of the strata are encountered, then soft band can be attacked first and the other once afterwards.
- If the roof rock is weak it is better to attack the central portion first, and then to rip the sides of the face/working.

Common machine parameters that play significant role in its cutting performance are:

- Cutting tool configuration
 - rake angle,
 - attack angle,
 - clearance angle and tip angle,
 - pick lacing,
 - type of pick,
 - number of picks,
 - tip material,
- Drum weight,
- Engine power,
- Loading options,
- Depth of cut
- Nominal cutting speed.

Table; Factors influencing Cutting performance

Main Factor	Variables	
Rock Parameters	Intact Properties	Cutability
		Cutting Wear
		Durability
	Mass Properties	Discontinuities Mixed Face Conditions Degree of Variation in Strata (along line of tunnel)
Machine Parameters	Environment	Water
		Tunnel Geometry
		In-situ Stresses
	Cutting Head	No. of Tools Tool Type Lacing Pattern
	Weight	Slewage - Lifting Forces
		Head Speed
		Head Power
	Operational Character	Rigidity of Machine Construction
		Profilling Guidance

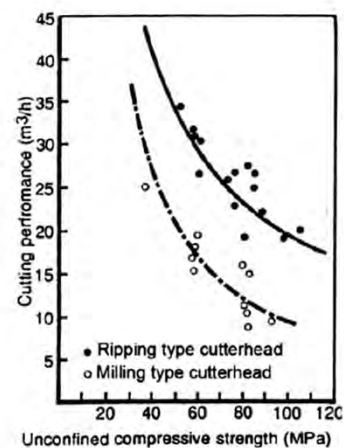


Fig. Comparison of cutting performance (Gehring, 1989)